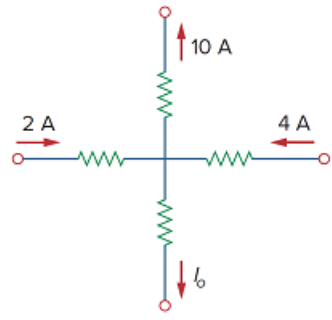


Problem

The current I_o of Fig. 2.64 is:

- (a) -4 A
- (b) -2 A
- (c) 4 A
- (d) 16 A

Figure 2.64:



Step-by-step solution

Step 1 of 4

Refer to Figure 2.64 in the textbook.

Redraw the circuit as shown in Figure 1.

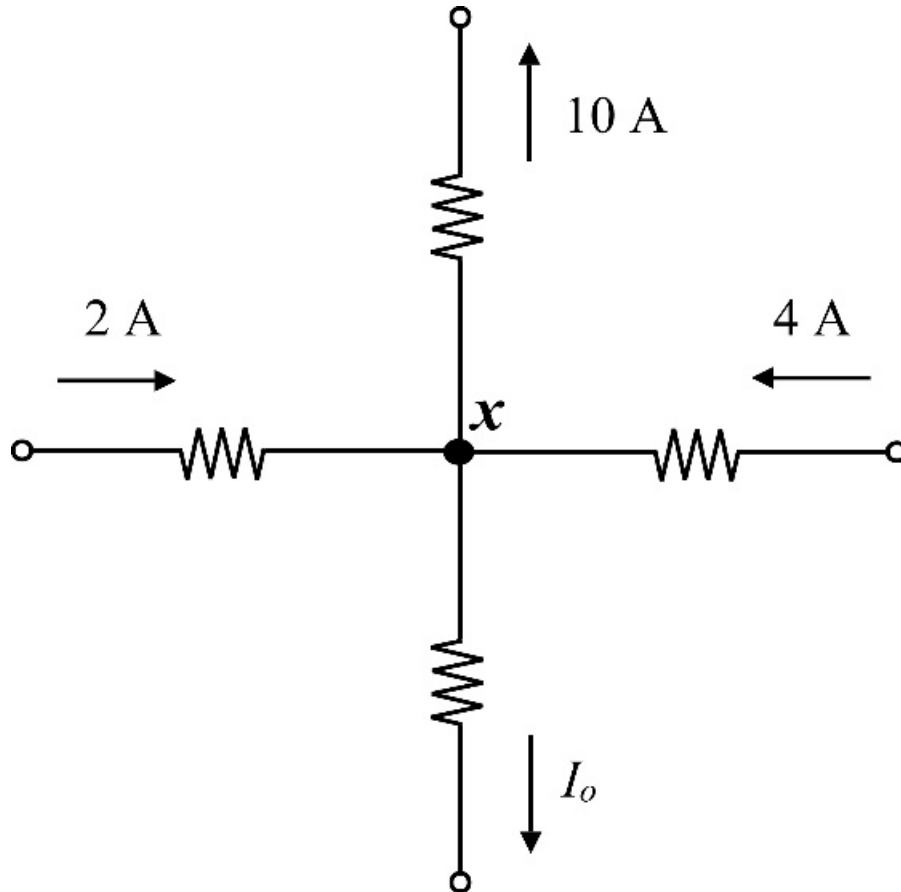


Figure 1

Step 2 of 4

Kirchhoff's current law states that the sum of all the currents entering a node is equal to the sum of all the currents leaving the node.

Apply Kirchhoff's current law at node x.

$$2 + 4 = 10 + I_o$$

$$I_o = 2 + 4 - 10$$

$$= -4 \text{ A}$$

Step 3 of 4

The current I_o cannot be -2 A , 4 A , or 16 A .

Therefore, the options (b), (c), and (d) are incorrect.

Step 4 of 4

The current I_o is -4 A .

Hence, the correct answer is option (a).